Gayathri Krishnamoorthy

Electrical Engineer| Computer Scientist

1630 NE Valley Road V102, Pullman, WA 99163

(509) 592-1317 | g.krishnamoorthy@wsu.edu | gayukrishna.github.io | gayathri-krishnamoorthy-6613b9a6

Education

Washington State University — *Pullman, WA*

Ph.D. in Electrical Engineering – *Major - Power Systems, Minor - Computer Science* — *January 2019 - Present*

- **Research** — *Transmission Distribution Co-simulation: Utility of Distribution Connected Renewable Resources in Providing Transmission System Support using Reinforcement Learning*

Washington State University — *Pullman, WA*

M.S. in Electrical Engineering – *Major - Power Systems, Minor - Computer Engineering* — *August 2016 - May 2018*

- **Research** — *An Iterative Co-simulation Framework for the Integrated Transmission and Distribution System Analysis*

Experience

Washington State University — *Pullman, WA*

Graduate Research Assistant — *August 2016 - Present*

- Developing a model-free power distribution system simulation for voltage control applications with probabilistic approaches and reinforcement learning techniques
- Designed and developed a transmission and distribution (T&D) co-simulation platform to understand the impacts of increasing renewable penetrations on power system operations
- Developed a hybrid T&D co-simulation platform that assists in secondary frequency regulation services to the power grid using distribution connected batteries
- Developed an optimization algorithm in a decentralized distribution market operation scenario to optimize the proposed market bids without violating system operating constraints

Pacific Northwest National Laboratory — *Seattle, WA*

Graduate Research Intern — *May. 2018 - Dec. 2018*

- Assisted in development of a transactive energy software agent, TESP, to optimize the dynamics of distributed energy resources in a household and bid the excess energy to provide monetary benefits to the costumers
- Performed testing of the transactive energy simulation platform with HELICS that integrated multiple power system market components on a Linux environment
- HVAC market participation simulations in day-ahead and real-time markets using Transactive energy simulation platform (TESP) with HELICS on a Linux environment
- Developed a C++ wrapper for MATPOWER software
- Developed a Docker environment for microgrid restoration application with FNCS

Projects

Model-free Distribution System Development using Reinforcement Learning — *2019 - Present*

- A model-free real-time distribution system voltage control using distributed energy resources is developed.
- Constrained policy gradient RL algorithm is implemented to control power distribution system voltages within operating limits using Torch. — *PSAT, OpenDSS,Python, Torch*

Real-time Control of Energy Storage Systems for Frequency Regulation of Power Grid — *2019 - Present*

- Applied game theoretic principals with deep RL to learn consumer willingness and flexibility in providing grid frequency regulation services
- Investigating the utilization of battery resources to provide combined active and reactive power support to mitigate the distribution grid violations — *MATLAB, OpenDSS, Python*

GMLC-TDC/ HELICS — *2018*

- Developer on HELICS, a coordinated effort by multiple National Labs to develop a platform that aids advanced power system operations.
- Performed scalability testing and wrapper development for multiple power system tools.
  — *C++, Python, Docker, MATLAB, Matpower*

Skills

Languages Python, C++, MATLAB, C, HTML, CSS, JavaScript

Tools Pytorch, Keras, TensorFlow, Git, Unix, OpenDSS, GridLAB-D, Matpower, PSAT, Docker

Awards and Honors

2018, 2019 **Paper Presentation,** IEEE Power and Energy Society General Meeting

2019 **Author,** Ph.D. Research Proposal to Pacific Northwest National Laboratory

2018 **2nd Place,** Graduate & Professional Student Association Scholarship Recipient for Research

2017 **2nd Place,** Best Student Paper Award at North American Power Symposium

2015 **Awardee,** Level 1 Cambridge English Language Assessment in Business English